

Create EKS Cluster in EKS Auto Mode

To Begin with the Lab

Summary of the Lab

In this step, a new **Amazon EKS** cluster was created in **Auto Mode**. The cluster name was entered, the Kubernetes version was selected, and the recommended cluster IAM role and node IAM role were created through **AWS IAM**. The cluster was then configured with the default **VPC** and the required subnets in **ap-south-1a** and **ap-south-1b**. After review, the cluster was created and allowed to move to the active state. This completed the first EKS setup step and prepared the cluster for later Kubernetes management and application deployment.

Understanding Amazon EKS Cluster Management

An **Amazon EKS** cluster is the place where Kubernetes workloads run, but the cluster itself still needs to be managed after creation. In EKS, you can manage the cluster through the **AWS Management Console** or through command line tools such as **AWS CLI** and **kubectl**. The console is useful for cluster overview, access entries, add-ons, node groups, networking, and monitoring, but it does not let you directly manage workloads like pods, services, ingress, logs, scaling, or ConfigMaps. Those workload-level tasks are handled with **kubectl**, which is why EKS administration usually uses both the console and the Kubernetes CLI.

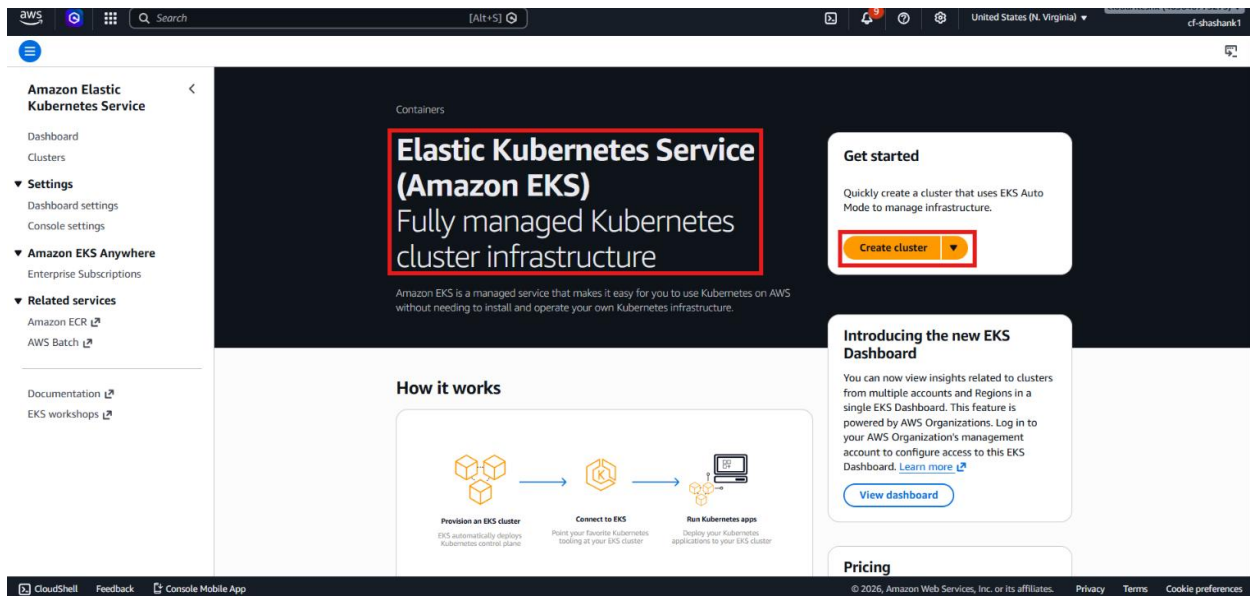
Example:

An administrator can open the EKS console to check cluster status, version, and access entries. But if the administrator wants to deploy an application pod or scale a deployment, that must be done with **kubectl** instead of the console. This separation keeps cluster management and workload management in the right tools.

Lab Steps

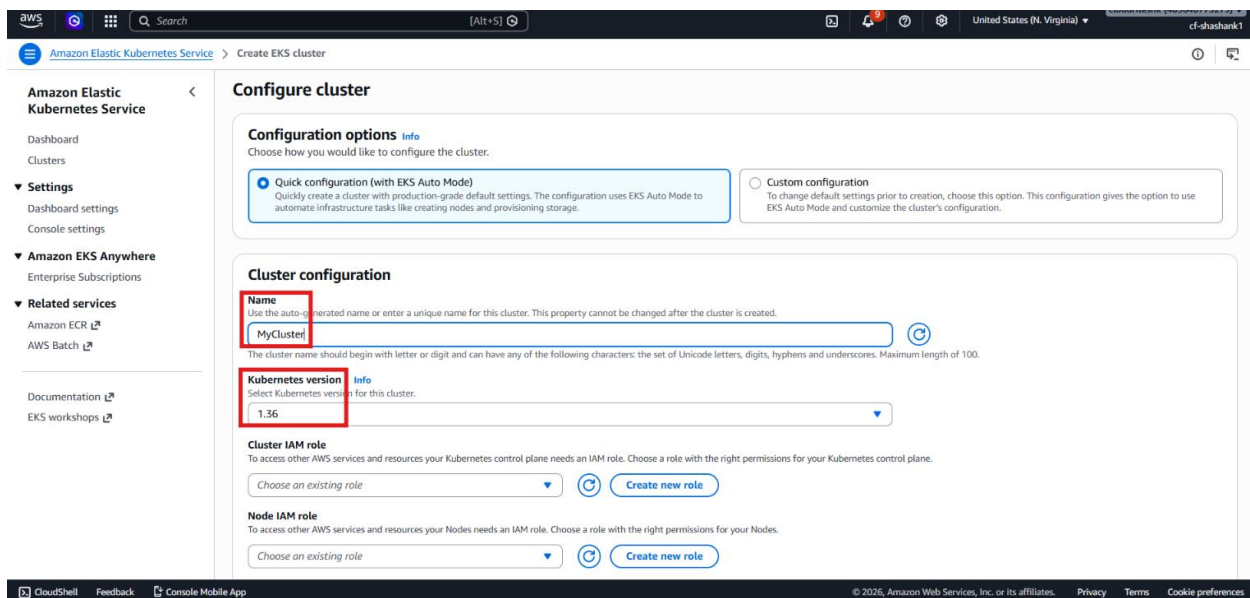
Step 1 – Open the EKS cluster creation page

- Sign in to the **AWS Management Console**.
- Open **Amazon EKS**.
- Click **Create cluster**.



Step 2 – Create the cluster in Auto Mode

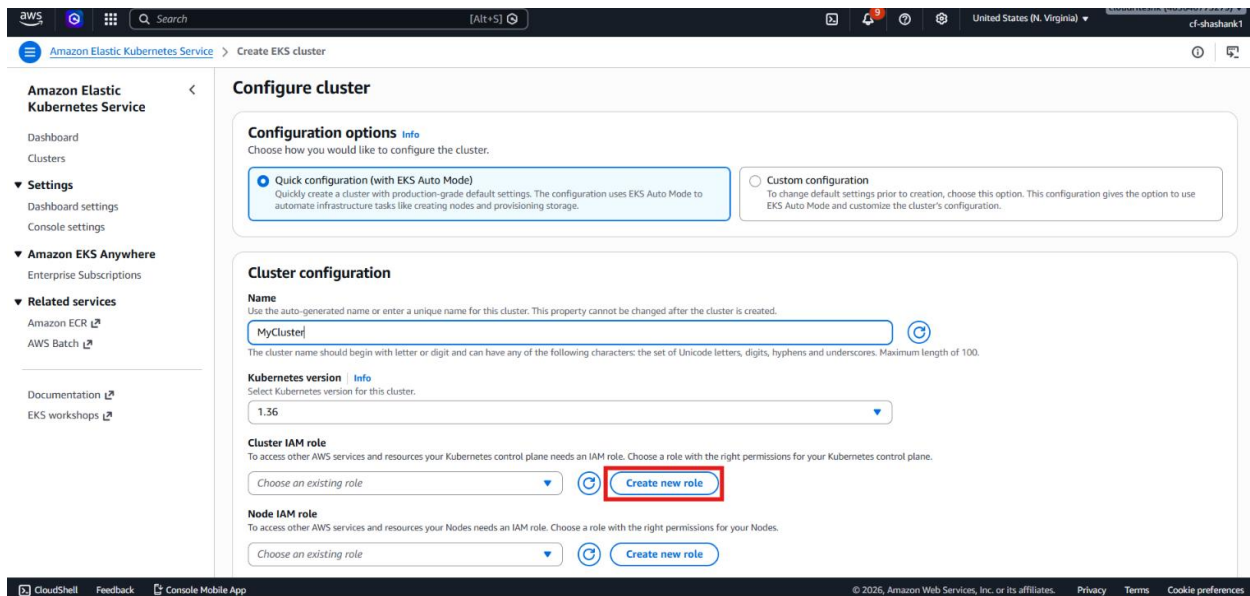
- Select **Auto Mode** for the cluster setup.
- Enter the cluster name: MyCluster
- Select the Kubernetes version.



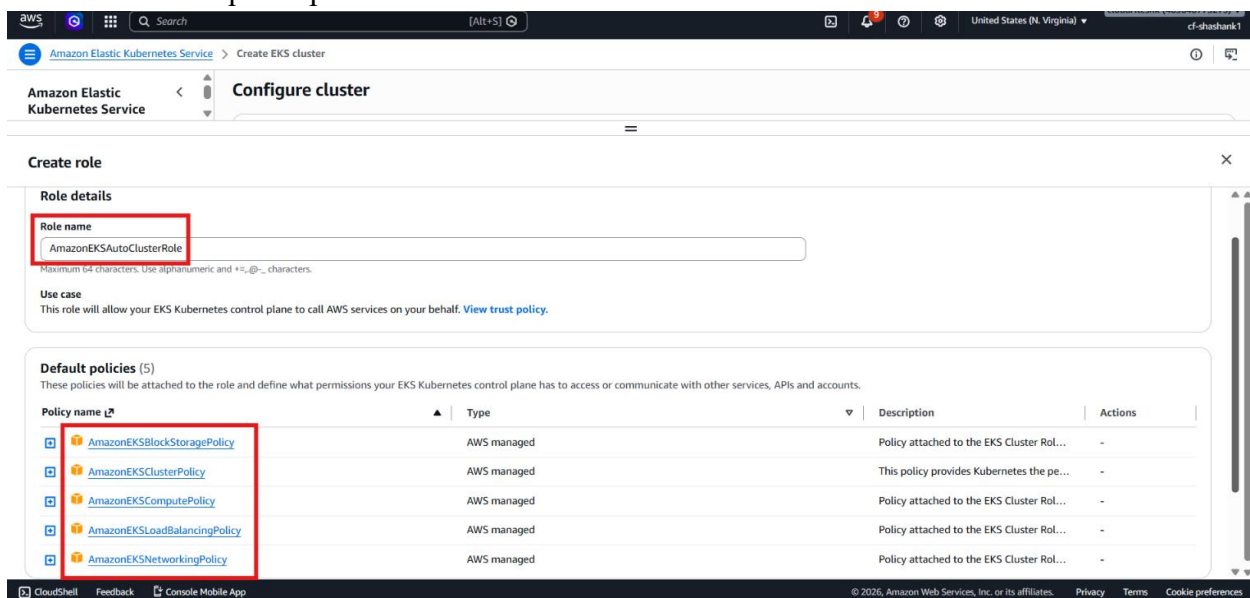
- Keep the default Auto Mode setup for the cluster.

Step 3 – Create the cluster IAM role

- Click **Create recommended role** for the **Cluster IAM role**.



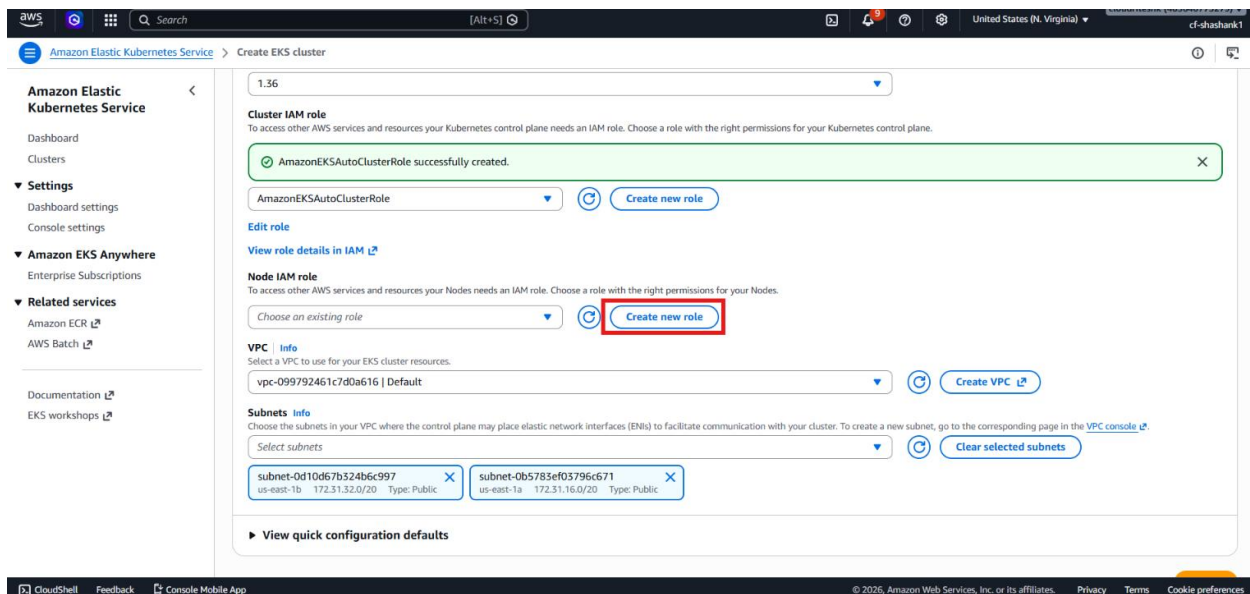
- Let **AWS IAM** open and create the role automatically.
- Attach the required permissions.



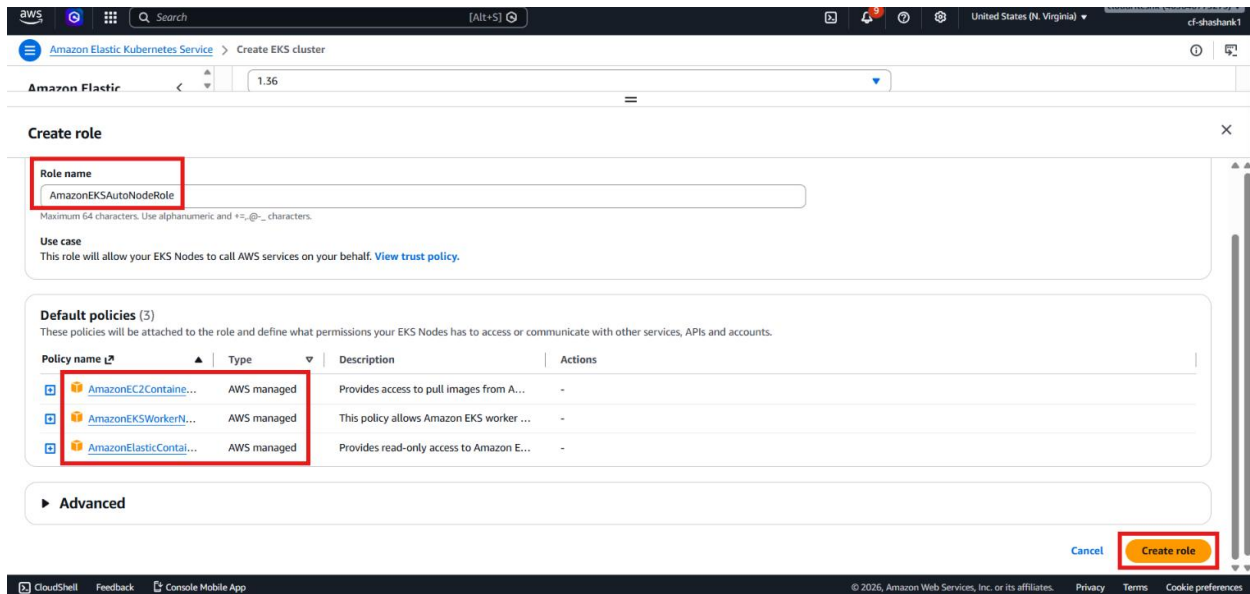
- Save and select the created cluster IAM role.

Step 4 – Create the node IAM role

- Click **Create recommended role** for the **Node IAM role**.

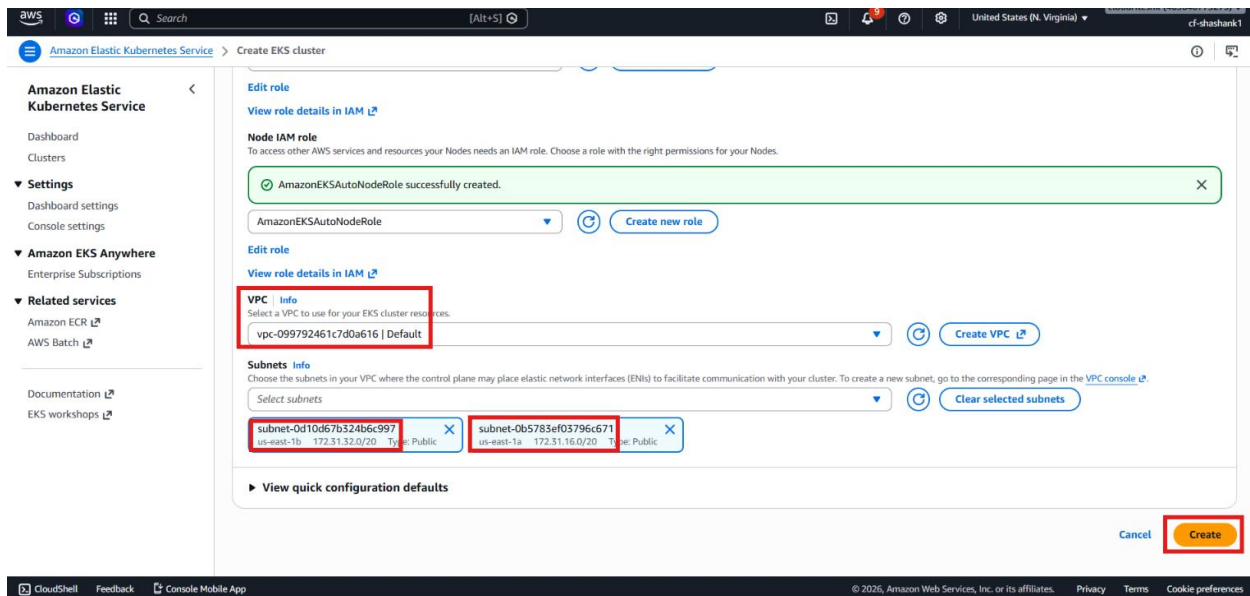


- Let AWS IAM create the node role automatically.
- Keep the default policy attachment.
- Select the created node IAM role.



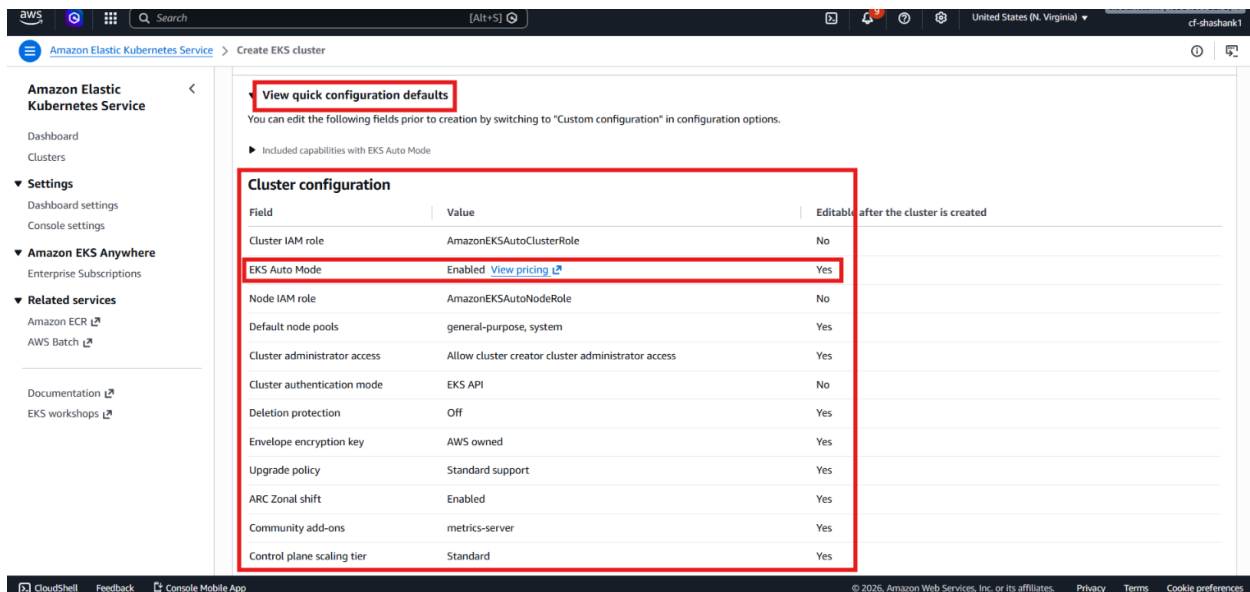
Step 5 – Configure VPC and subnets

- Use the default VPC.
- Select the required subnets in **ap-south-1a** and **ap-south-1b** if you are working in **Mumbai** region but it changes to respective clusters.
- I am working in **N.Virginia** so that's why you can see **us-east-1a** and **us-east-1b**.
- Review the networking details shown for the cluster.



Step 6 – Review the cluster settings

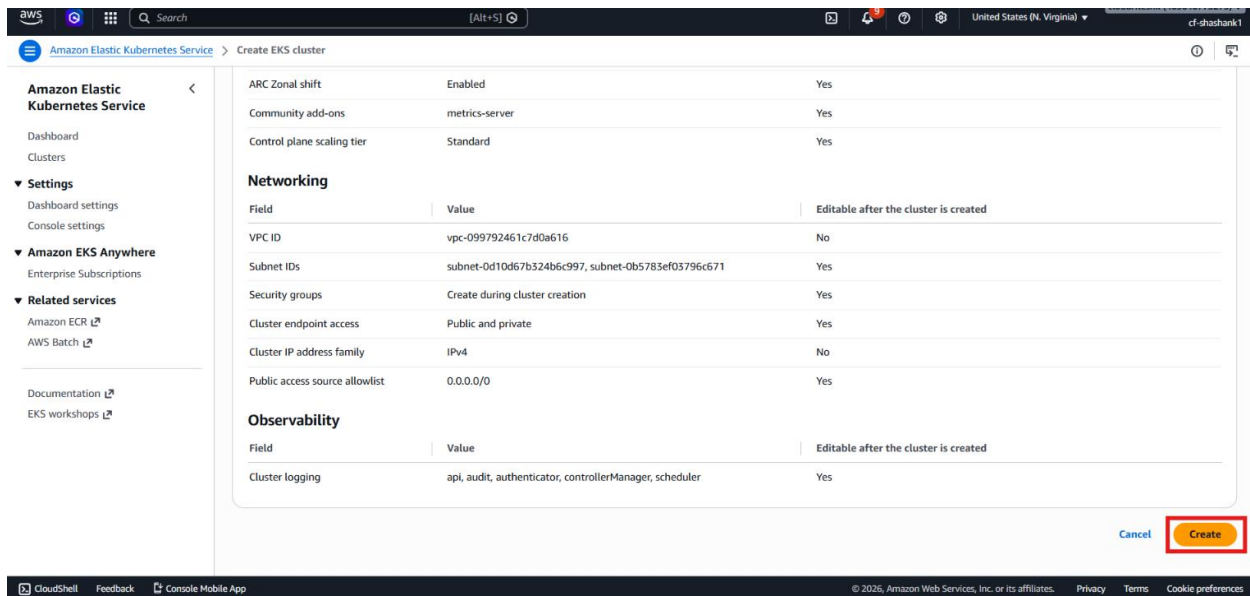
- Confirm that EKS Auto Mode is enabled.



- Review the remaining default settings.
- Verify that the settings shown are the expected Auto Mode defaults.

Step 7 – Create the cluster

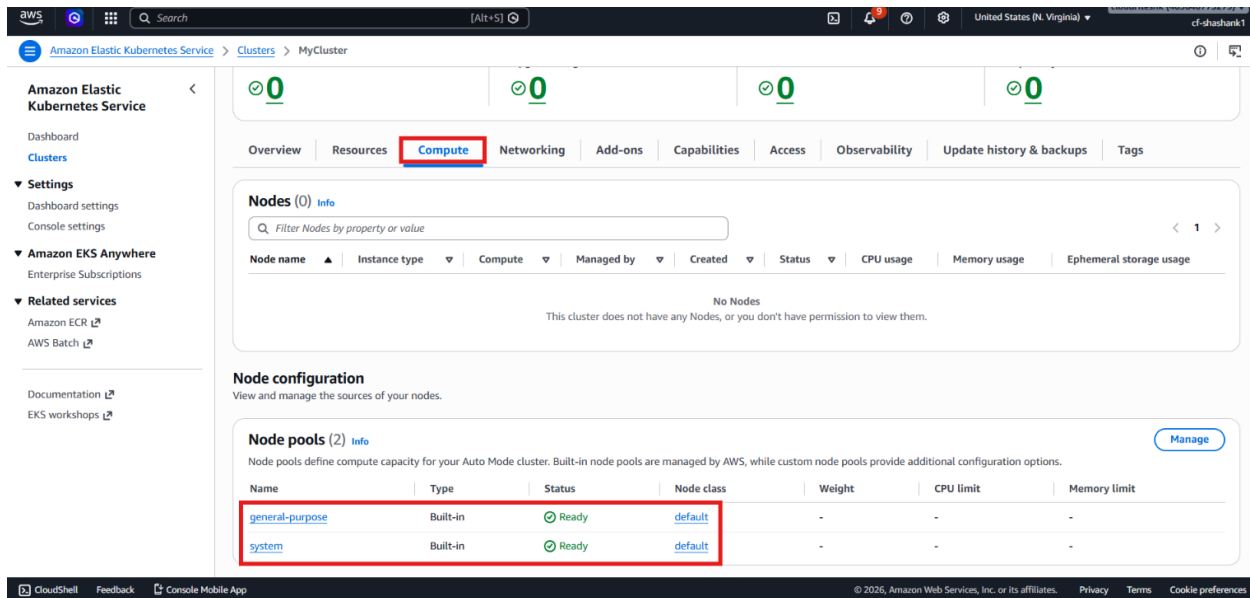
- Click **Create**.



- Wait for the cluster creation to finish.
- Confirm that the cluster status becomes active.

Step 8 – Review cluster management options

- Open the cluster details.
- Check the cluster overview, access management, add-ons, networking, and monitoring sections.



- Note that pods, services, ingress, logs, scaling, and ConfigMaps are managed with **kubectl**, not from the console.

Verification

- Verified that the **Amazon EKS** cluster was created successfully.

- Verified that **EKS Auto Mode** was enabled for the cluster.
- Verified that the cluster IAM role and node IAM role were created and attached.
- Verified that the cluster used the selected **VPC** and subnets.
- Verified that the cluster status became active.
- Verified that the console can manage cluster settings, while workload operations require **kubectl**.